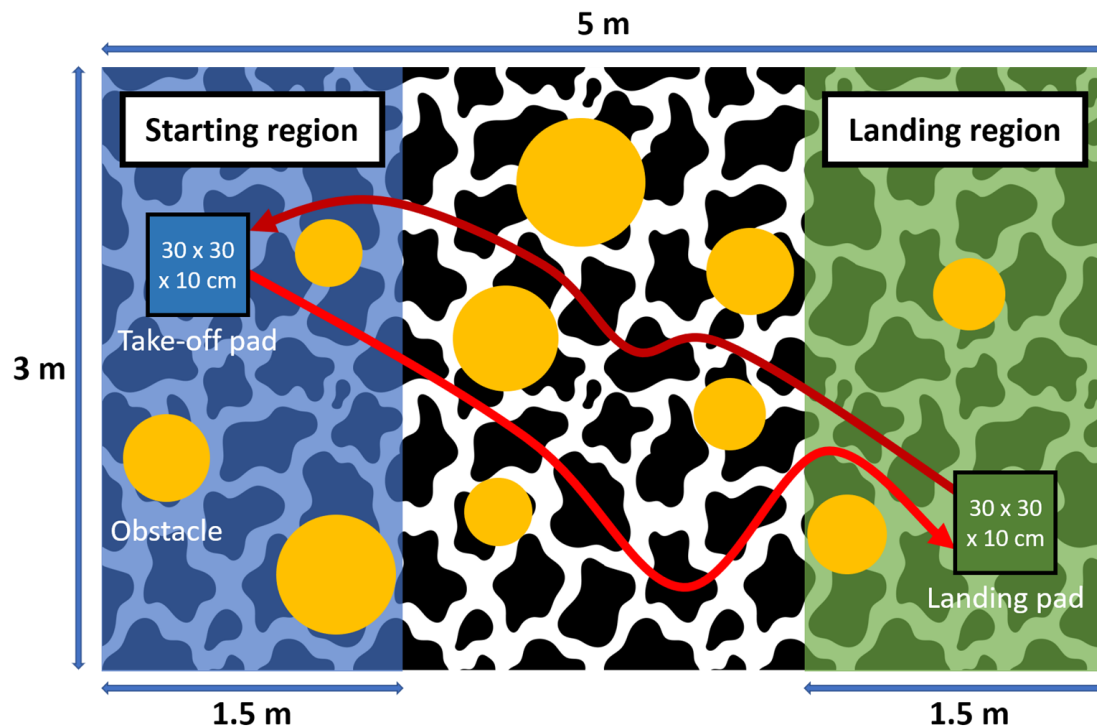




Crazy Practical

Introduction

Task Setup

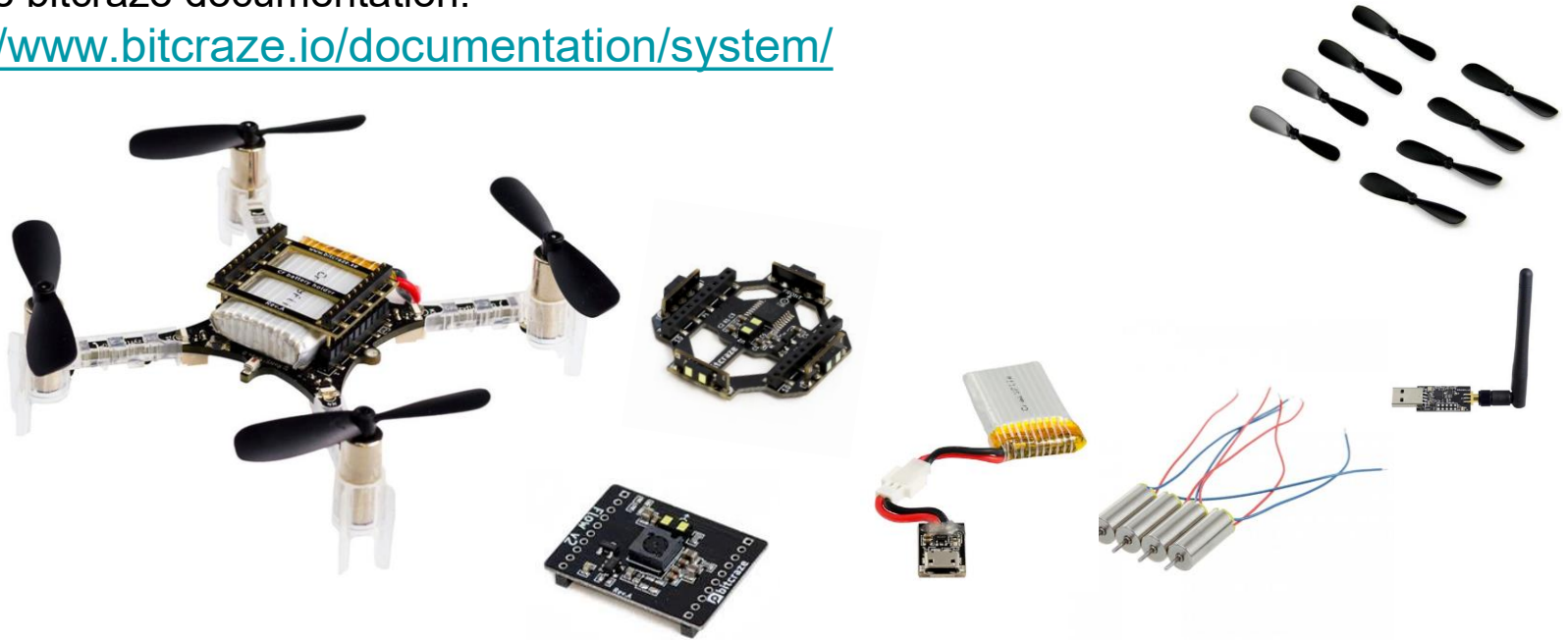


- Arena: pattern tiles (3 x 5 m area)
- Platform: 2 x carton box (~ 30 x 30 x 10 cm³ volume)
- Drone: initially placed on the take-off pad
- Task: navigation and obstacle avoidance
 - Estimate position from flow deck
 - Use multi-ranger to detect and avoid obstacles
 - Use z-ranger to detect and land on the platform

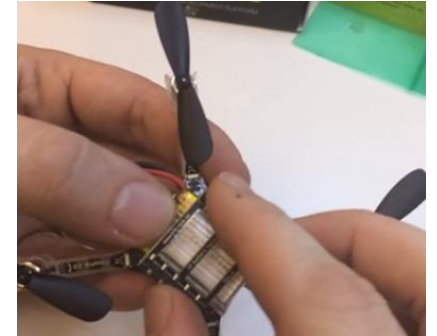
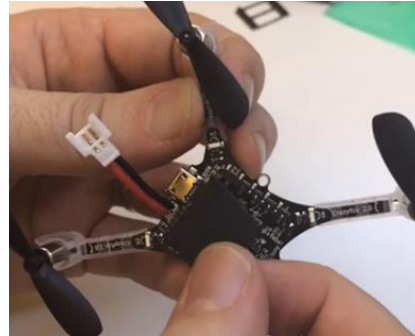
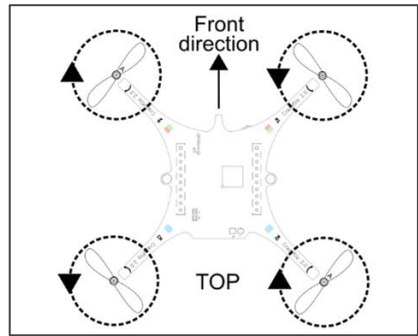
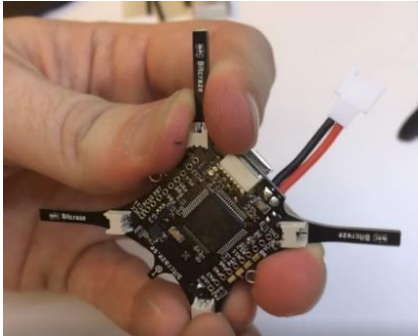
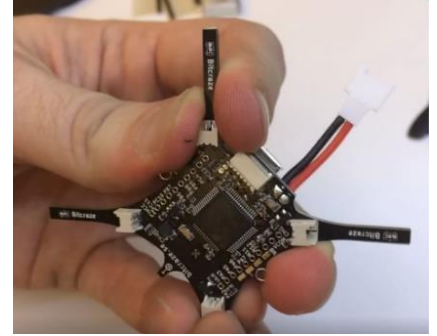
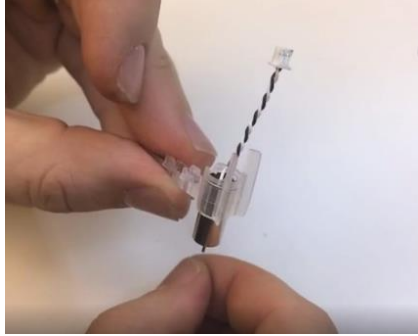
1. Unpacking

Visit the bitcraze documentation:

<https://www.bitcraze.io/documentation/system/>



2. Assembling



3. Installing the Python library

- **cflib: Crazyflie python library**

- API written in Python
- used to communicate with the Crazyflie
- used to control the Crazyflie
- used by [cfclient](#), the Crazyflie PC client

- Visit the doc here: <https://www.bitcraze.io/documentation/repository/crazyflie-lib-python/master/>

3.1 Installing Python

The image shows a web browser window displaying the Python 3.12.4 download page. The page includes links to online documentation, PEP 693, GitHub issues, and sponsors. Below the resources, there is a table of files for download. A yellow arrow points from the text 'Make sure to add python.exe to PATH' to the 'Add python.exe to PATH' checkbox in the Windows installer setup window.

For more details on the changes to Python 3.12, see [What's new in Python 3.12](#).

More resources

- Online Documentation.
- PEP 693, the Python 3.12 Release Schedule.
- Report bugs via [GitHub Issues](#).
- Help fund Python directly or via [GitHub Sponsors](#), and support the Python community.

[Full Changelog](#)

Files

Version	Operating System	Description	MD5 Sum	File Size	GPG
Gzipped source tarball	Source release		ead819dab6d165937138daa9e51ccb54	26.0 MB	SIG
XZ compressed source tarball	Source release		d68f25193eec491eb54bc2ea664a05bd	19.7 MB	SIG
macOS 64-bit universal2 installer	macOS	for macOS 10.9 and later	b6de6aea008605f5d4096014c2ad3c43	44.0 MB	SIG
Windows installer (64-bit)	Windows	Recommended	f3df1be26cc7cbd8252ab5632b62d740	25.5 MB	SIG
Windows installer (ARM64)	Windows	Experimental	f3c2064f11c5f4eee475928a0fc62199	24.8 MB	SIG
Windows embeddable package (64-bit)	Windows		8db759b337ac4f6966f52b3662c05dd7	10.6 MB	SIG
Windows embeddable package (32-bit)	Windows		19691145551a41114b32a556bb2bcb89	9.4 MB	SIG
Windows embeddable package (ARM64)	Windows		0a863fd2485b3057a2eea108f1252160	9.8 MB	SIG
Windows installer (32-bit)	Windows		d9c98b529889aba04ca5ec1c6b5f986f	24.3 MB	SIG

Python 3.12.4 (64-bit) Setup

Install Python 3.12.4 (64-bit)

Select Install Now to install Python with default settings, or choose Customize to enable or disable features.

→ **Install Now**
C:\Users\William Stewart\AppData\Local\Programs\Python\Python312

Includes IDLE, pip and documentation
Creates shortcuts and file associations

→ **Customize installation**
Choose location and features

☐ Use admin privileges when installing py.exe

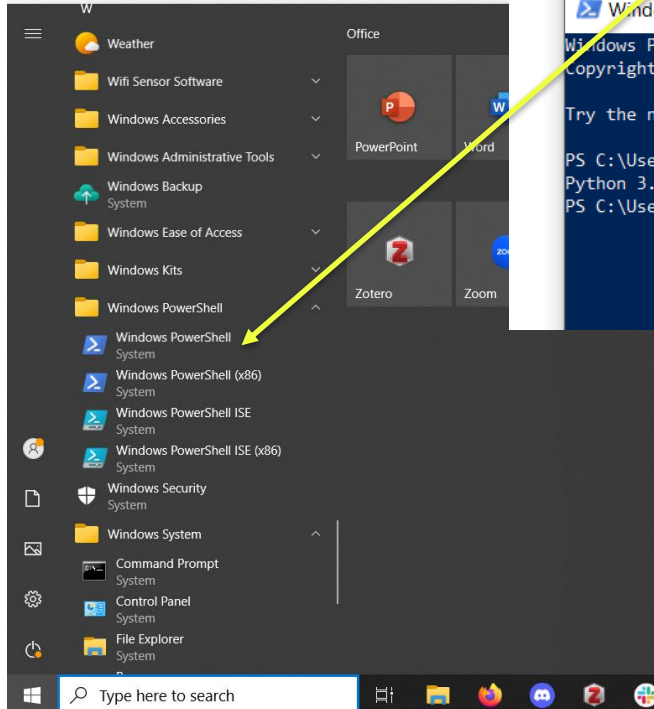
☒ Add python.exe to PATH

Cancel

Make sure to add python.exe to PATH

3.2 Running Python

Open Windows PowerShell, run the command below.



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\William Stewart> python --version
Python 3.12.4
PS C:\Users\William Stewart>
```

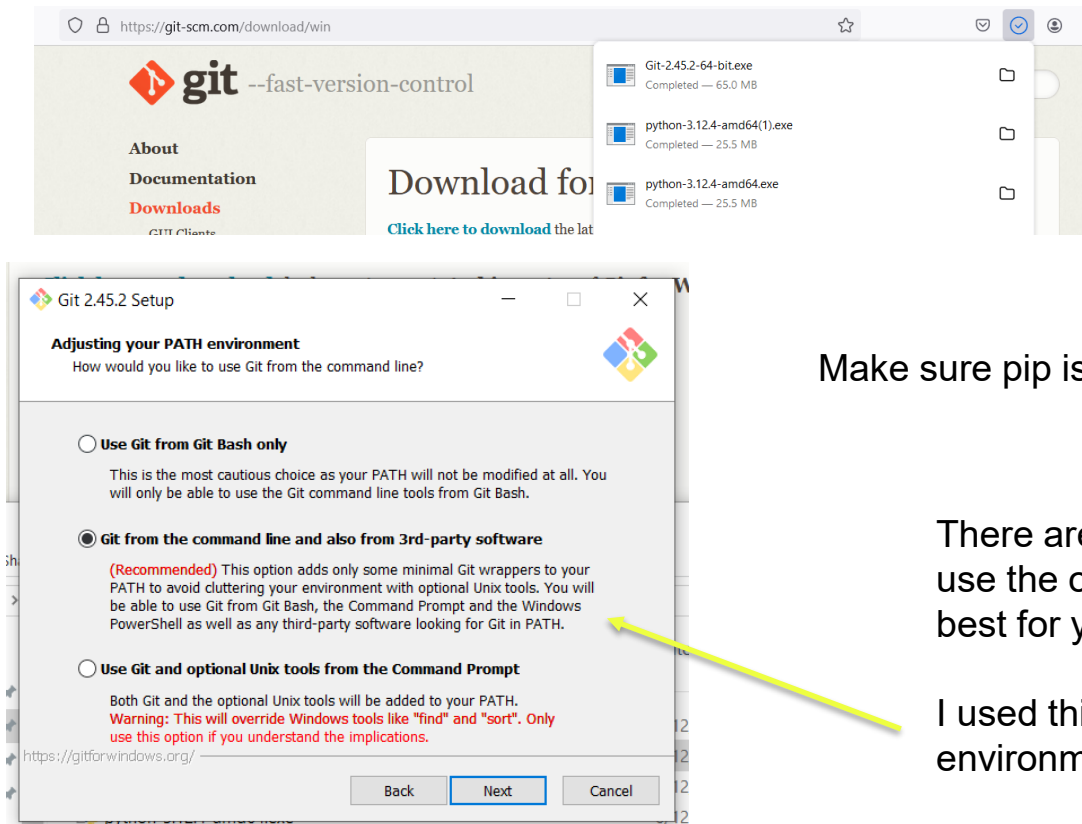
Make sure pip is installed and upgraded.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\William Stewart> python --version
Python 3.12.4
PS C:\Users\William Stewart> pip --version
pip 24.0 from C:\Users\William Stewart\AppData\Local\Programs\Python\Python312\Lib\site-packages\pip (python 3.12)
PS C:\Users\William Stewart> pip3 install --upgrade pip
Requirement already satisfied: pip in c:\users\william stewart\appdata\local\programs\python\python312\lib\site-packages
(24.0)
PS C:\Users\William Stewart>
```

3.3 Installing GIT



Make sure pip is installed and upgraded.

There are lots of git options, use the ones you think work best for you.

I used this for the PATH environmental variable.

4. Setting up the radio interface

- Make sure you have the usb permission for the radio interface
- Configure the crazyradio and set your crazyflie address
- Remember to change the uri in the files of your algorithm



4.1 Install Zadig

<https://www.bitcraze.io/documentation/repository/crazyradio-firmware/master/building/usbwindows/>

Blog Products Buy Support Document

USB driver on Windows

[repository overview](#) / [crazyradio-firmware](#) / [master](#) / Building / Installing USB driver on Windows

Home

Building and Flashing

Build and flash instructions

crazyradio firmware upgrade

Installing USB driver on Windows

Explanation Functionalities

protocol of the Crazyradio

Most Bitcraze products using USB communication, uses the libusb driver. This page explains how to install the USB driver on Windows without having to develop custom drivers. This page explains how to install the USB driver on Windows without having to develop custom drivers. This page explains how to install the USB driver on Windows without having to develop custom drivers.

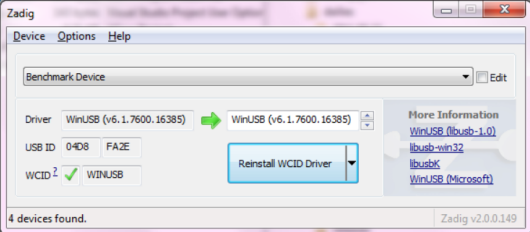
Install the driver

First of all download Zadig from its website: <http://zadig.akeo.ie/>.

Plug the Bitcraze device in the PC. Windows should display some installation wizard and closes itself. On windows 7 you have to close the install failed wizard.

<https://zadig.akeo.ie>

USB driver installation made easy



Zadig is a Windows application that installs generic USB drivers, such as [WinUSB](#), [libusb-win32](#), [libusbK](#), or [WinUSB \(Microsoft\)](#), to help you access USB devices.

It can be especially useful for cases where:

- you want to access a device using a libusb-based application
- you want to upgrade a generic USB driver
- you want to access a device using WinUSB

Note: "libusb-based" above means an application that uses either [libusb](#), [libusb-win32](#) or [libusbK](#).

Download

Updated 2023.03.01:

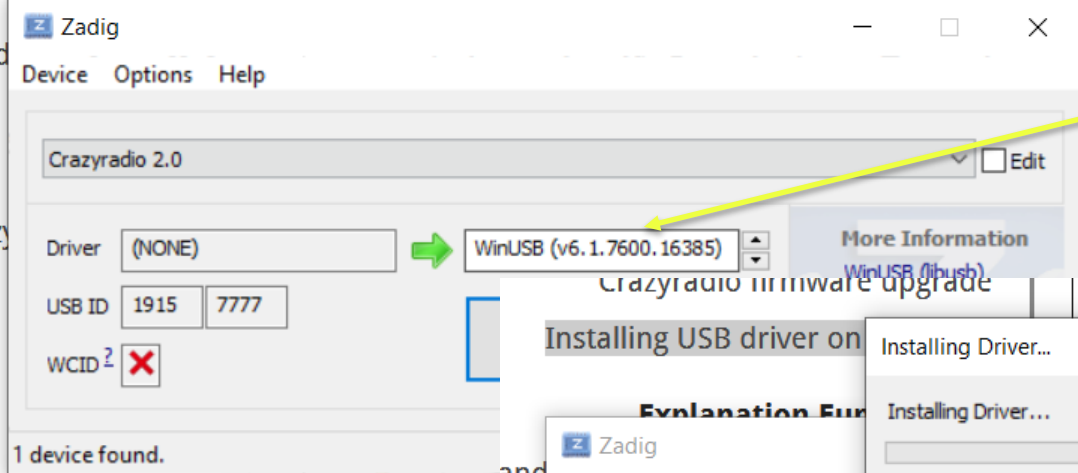
- [Zadig 2.8](#) (5.0 MB)
- [Other versions](#)

System Requirements:

Installing USB driver on Windows

Plug the Bitcraze device in the PC. Windows
and closes itself. On windows 7 you have to

Explanation Functionalities



I had to replace the driver cause I did the wrong one

Installing USB driver on

Installing Driver...

Installing Driver...

First of all download Zadig from its website

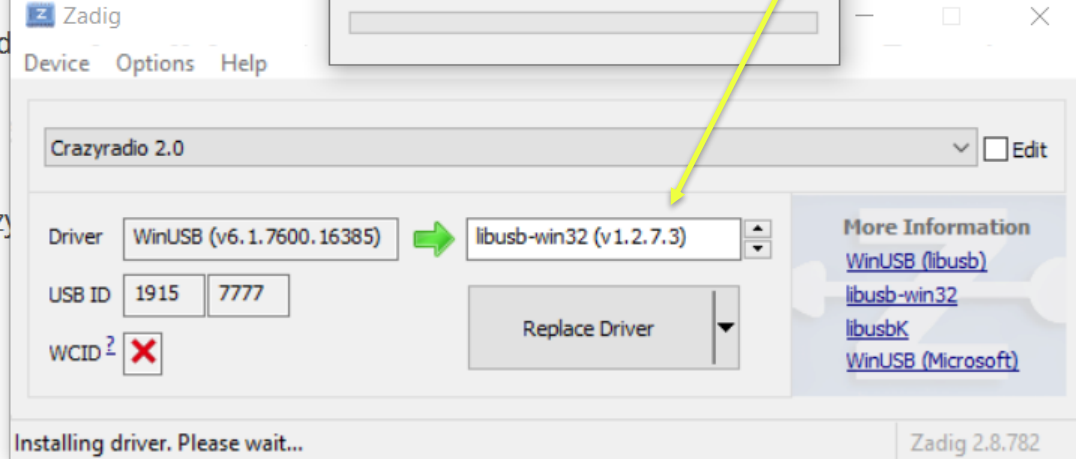
device in the PC. Window

on windows 7 you have t

and

Explanation Fur

razy



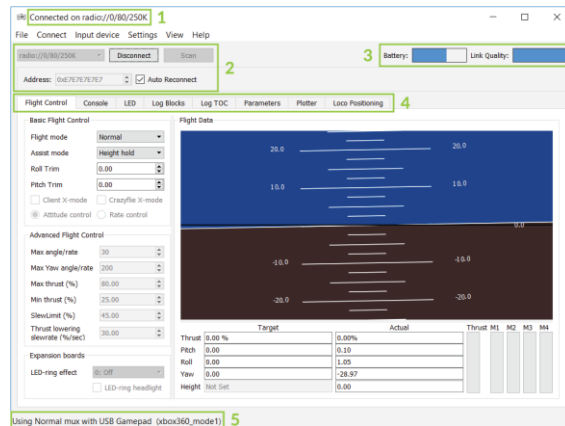
5. Installing the Crazyflie client

- **cfclient: Crazyflie PC Client**

- written in Python
- used to flash, change settings, control, and visualize logs of the Crazyflie
- implements the user interface

- Visit the doc here:

- <https://www.bitcraze.io/documentation/repository/crazyflie-clients-python/master/>
- <https://github.com/bitcraze/crazyflie-clients-python>



5.1 Installation

https://www.bitcraze.io/documentation/repository/crazyflie-clients-python/master/installation/install/



Installing from latest release

If you plan to use the client to control the Crazyflie, we highly recommend you to install the latest release as it is a well tested and stable. Please see next section.

On the other hand, if you intend to do development work on the client and modify the source code, please see [source](#) below.

From Pypi (Windows, Mac, Linux, ..., with python3)

Each release of the client is pushed to the [pypi repository](#), so it can be installed with pip:

```
pip3 install cfclient
```

Start application by typing

```
git version 2.43.2.windows.1
PS C:\Users\William Stewart> pip3 install cfclient
Collecting cfclient
  Downloading cfclient-2024.3-py3-none-any.whl.metadata (2.4 kB)
Collecting cflib<=0.1.25.1 (from cfclient)
  Downloading cflib-0.1.25.1-py2.py3-none-any.whl.metadata
Collecting setuptools (from cfclient)
  Downloading setuptools-70.0.0-py3-none-any.whl.metadata
Collecting appdirs<=1.4.0 (from cfclient)
  Downloading appdirs-1.4.4-py2.py3-none-any.whl.metadata
Collecting pyzmq<=25.0 (from cfclient)
  Downloading pyzmq-25.1.2-cp312-cp312-win_amd64.whl.metadata
Collecting pyqtgraph<=0.13 (from cfclient)
  Downloading pyqtgraph-0.13.7-py3-none-any.whl.metadata
Collecting PyYAML<=6.0.1 (from cfclient)
```

```
cfclient
```

```
Successfully installed PyQt6-Qt6-6.7.1 PyQt6-sip-13.6.0 PyYAML-6.0.1 appdirs-1.4.4 cflib-0.1.25.1 cfclient-2024.3 kiwisolver-1.4.5 libusb-package-1.0.26.1 numpy-1.26.4 pyqtgraph-0.13.7 pysdl2-0.9.16 pysdl2-dll-2.24.0 pyserial-3.5 pyusb-1.2.1 pyzmq-25.1.2
PS C:\Users\William Stewart> cfclient
```

```
UserWarning: Using SDL2 binaries from pysdl2-dll 2.24.0
```

Need to find your crazyflie here. You've been given an address, it's on a sticky note.
Try not to change it. Definitely don't forget it.

Not connected

File Connect Input device Settings View Help Themes

Select an interface Battery: volts Link Quality:

Address:

Flight Control

Basic Flight Control

Flight mode

Assist mode

Roll Trim

Pitch Trim

Advanced Flight Control

Max angle/rate

Max Yaw angle/rate

Max thrust (%)

Min thrust (%)

SlewLimit (%)

Thrust lowering

Flight Data



No input-device found, insert one to fly.

Connected

This window shakes when you connect to your crazyflie!

Connected on radio://0/88/2M/E7E7E7E0

File Connect Input device Settings View Help Themes

radio://0/88/2M/E7E7E7E0 Disconnect Scan Emergency stop Battery:  3.832 volts Link Quality: 

Address: 0xE7E7E7E0

Flight Control

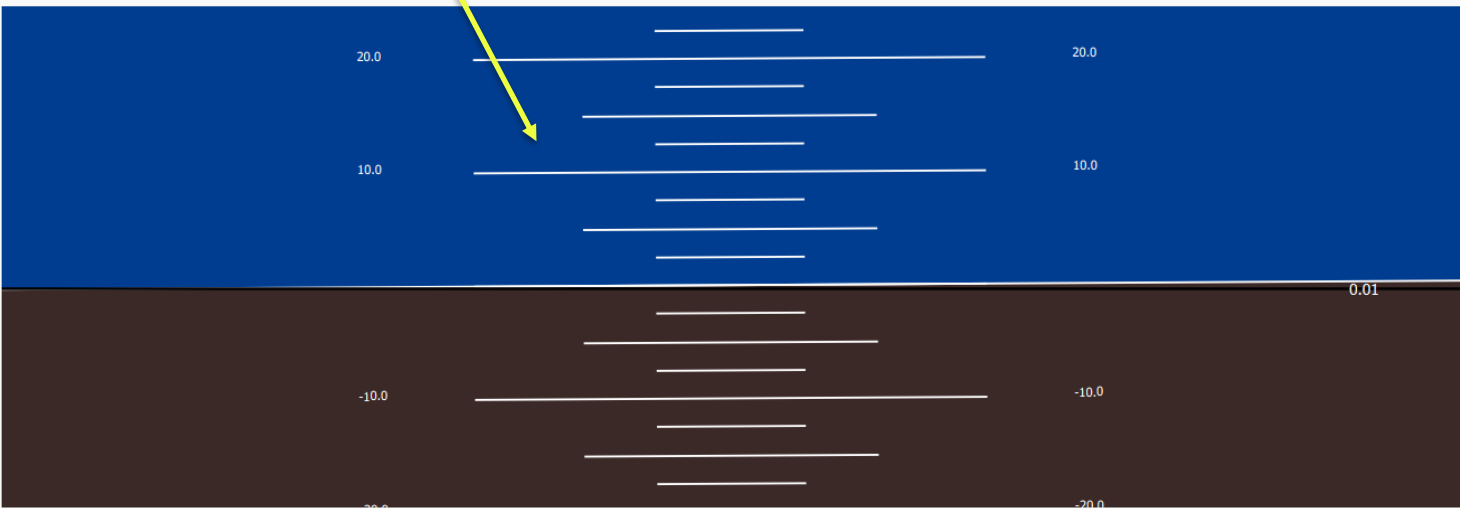
Basic Flight Control

Flight mode: Normal
Assist mode: Hover
Roll Trim: 0.00
Pitch Trim: 0.00

Advanced Flight Control

Max angle/rate: 30
Max Yaw angle/rate: 200
Max thrust (%): 80
Min thrust (%): 25
SlewLimit (%): 45
Thrust lowering: 20

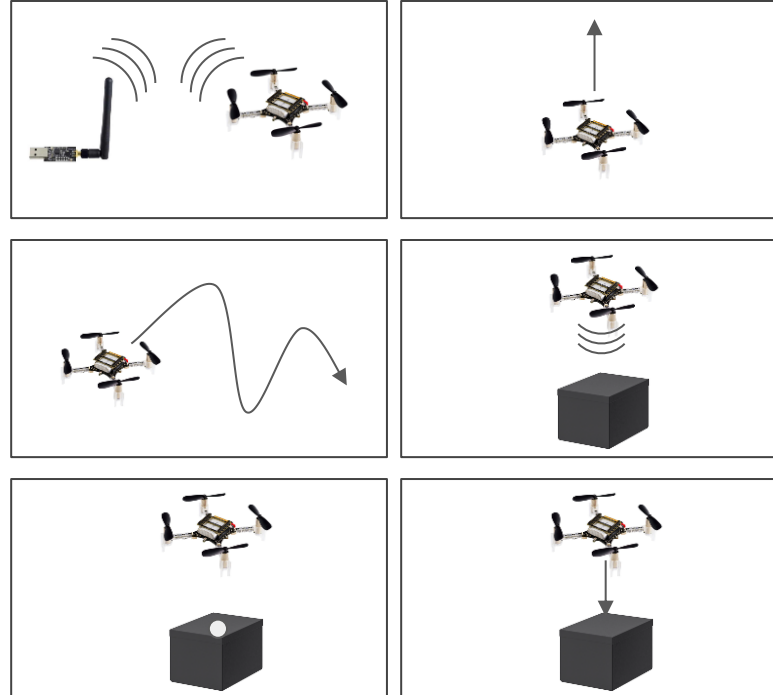
Flight Data



No input-device found, insert one to fly.

6. Coding your algorithm

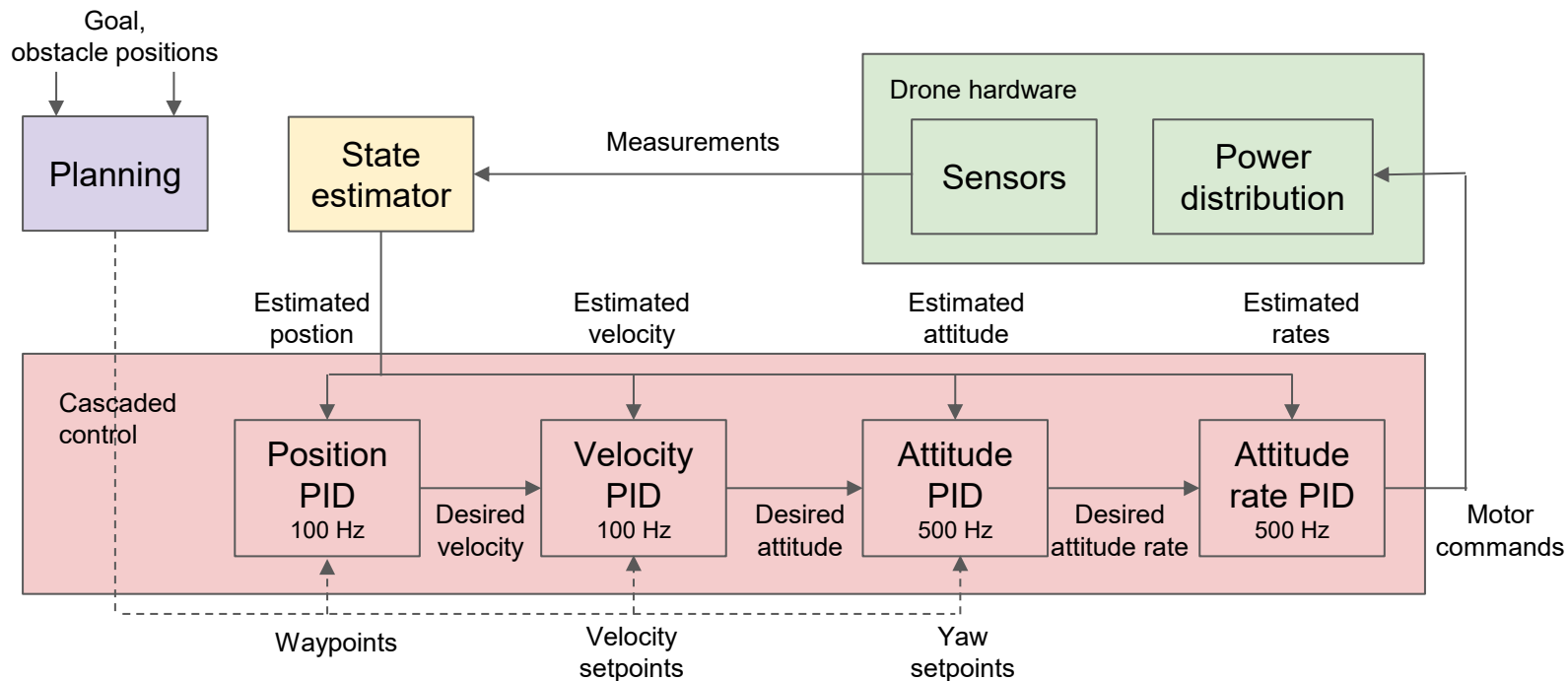
- Connect to the crazyflie
- Take off
- Explore the arena
- Avoid collisions
- Detect the box
- Estimate the box center
- Land on it



- To run code:
 - Use powershell
 - Cd ./directory_with_script
 - Must be disconnected from cfclient

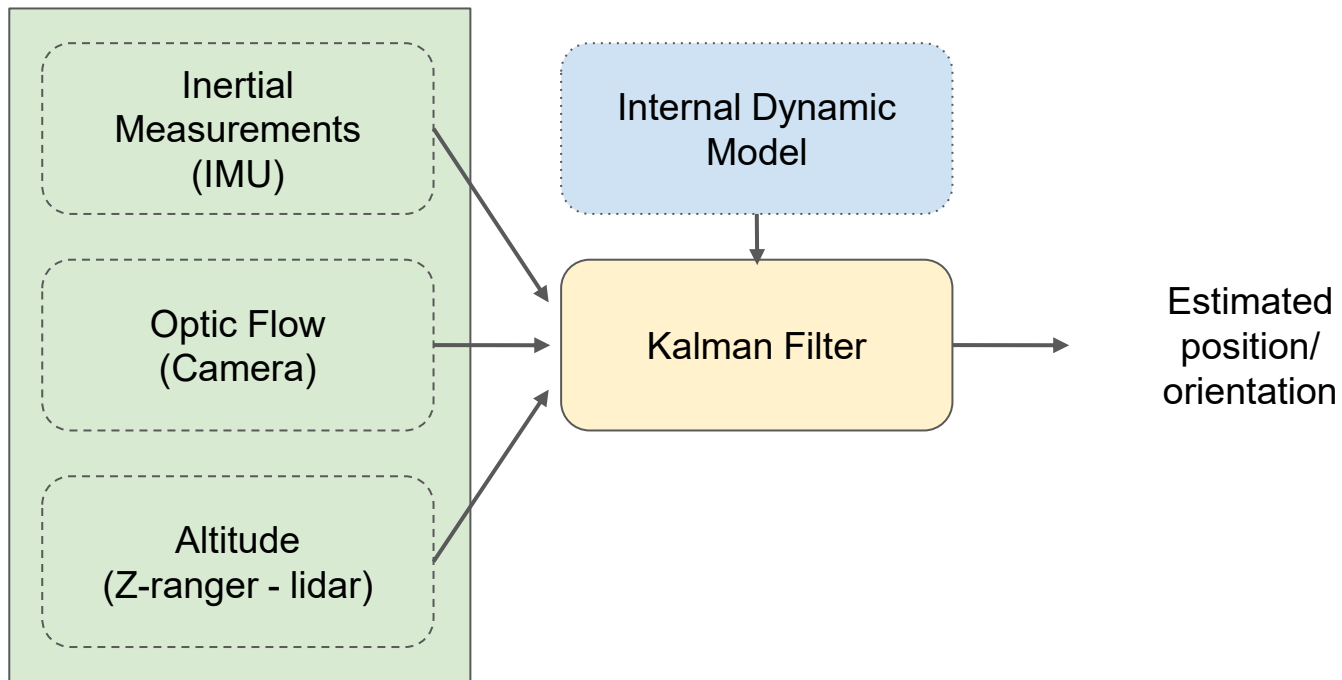
```
PS C:\Users\William Stewart> cd .\Dropbox\AAA_StonyBrookUniversity\Aerial-Robotics_Course\Fall-2024\crazy-Practical
PS C:\Users\William Stewart\Dropbox\AAA_StonyBrookUniversity\Aerial-Robotics_Course\Fall-2024\crazy-Practical> python example_code.py
Taking off!
Moving up 0.2m
Moving down 0.2m
Landing!
PS C:\Users\William Stewart\Dropbox\AAA_StonyBrookUniversity\Aerial-Robotics_Course\Fall-2024\crazy-Practical> █
```

Crazyflie software architecture

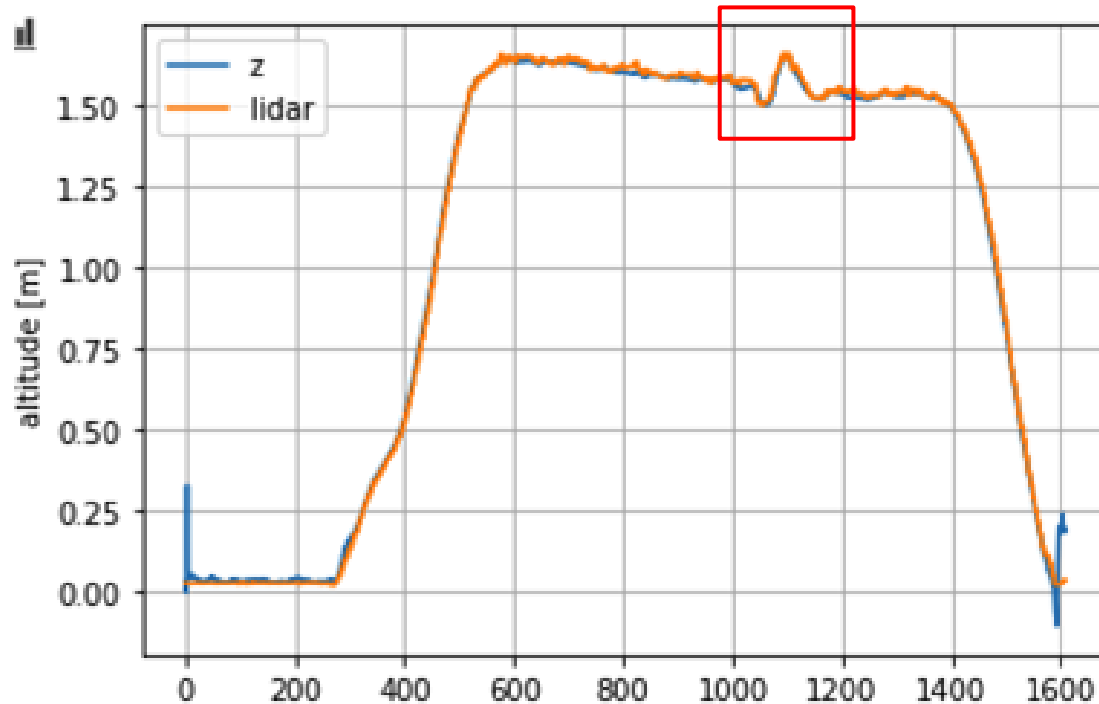


<https://www.bitcraze.io/documentation/repository/crazyflie-firmware/master/functional-areas/sensor-to-control/controllers/>

Altitude estimation



Forward flight at constant altitude



What's next?

- Do the installation
 - <https://github.com/dronecourse-epfl/crazy-practical-2022>
- Start coding
- Charge battery(s)